

Reconstruction of Quantum Illumination with Gaussian States

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Final Presentation for Exploration Project in Physics 2 (PHYS0352)

Contents

- Status
- Quantum illumination theory
 - OPA(optical parametric amplifier)
 - FF-SFG receiver
 - Performance : ROC
- Calculation Gaussian quantum illumination
 - Normal mode decomposition of Gaussian states
 - Reconstruction Classical illumination (for the Coherent state transmitter)
 - Reconstruction of Quantum illumination (for the TMSV transmitter)
- Results
- Discussion & Further research



QuTiP
Quantum Toolbox in Python

- Calculate Gaussian QI
 - Gaussian single mode
 - QCB, QBB, ROC



WOLFRAM
MATHEMATICA 13

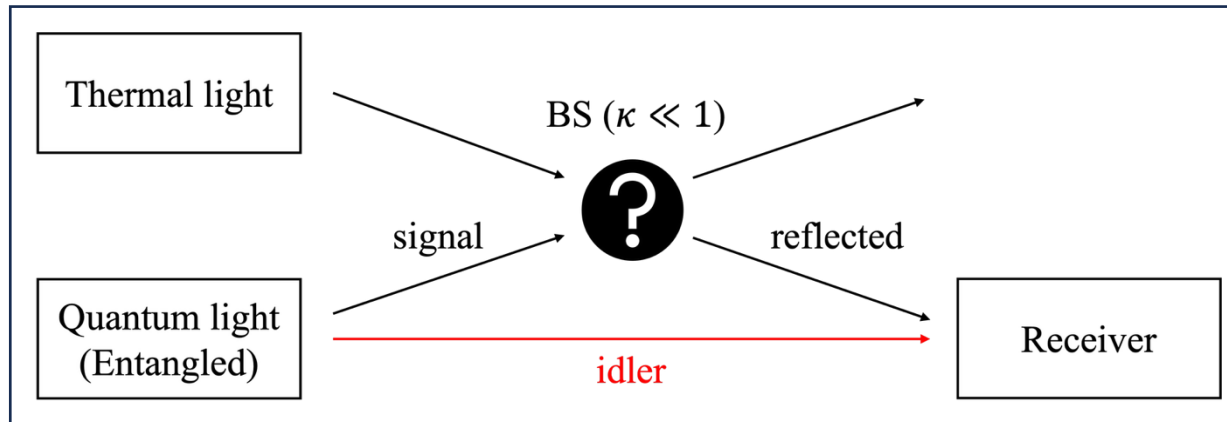
- Calculate Gaussian QI
 - Coherent state, TMSV
 - QCB, QBB

- Quantum illumination theory
 - Practical receiver for quantum illumination
 - The ultimate receiver: the FF-SFG receiver
 - Performances: ROC for quantum illumination
 - Criticalities and limitations
- Experiments on quantum illumination

Recall : Quantum illumination

- Quantum illumination

: Quantum target detection protocol



case	decision	real	
1	H_0	H_0	correct
2	H_1	H_0	false alarm error
3	H_1	H_1	correct
4	H_0	H_1	miss error

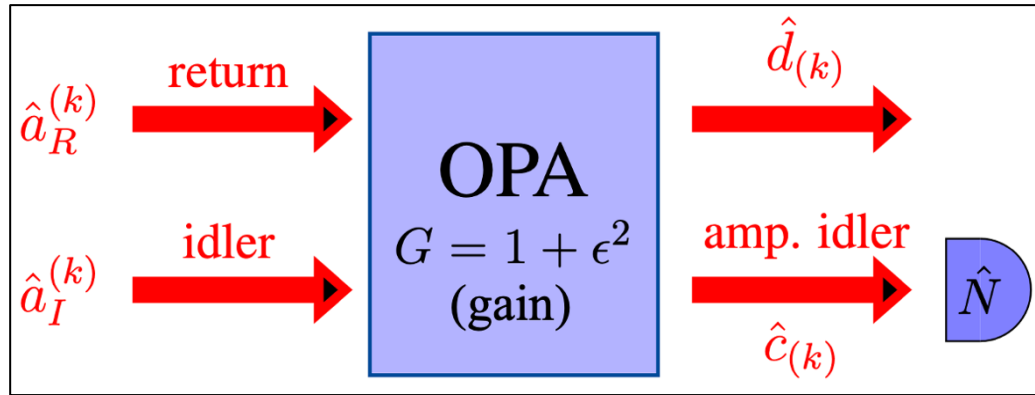
Choose E_i ($i = 0,1$) that minimize $P_e = \omega_0 \text{tr}(E_1 \rho_0) + \omega_1 \text{tr}(E_0 \rho_1)$

$$P_e = \frac{1 - \text{tr}|\omega_1 \rho_1^{\otimes M} - \omega_0 \rho_0^{\otimes M}|}{2} \quad : \text{Helstrom bound}$$

$$P_e \leq \frac{1}{2} e^{-M \xi_{QC}} \leq \frac{1}{2} e^{-M \xi_{QB}} \quad \begin{aligned} \xi_{QB} &= -\log[\text{tr}(\sqrt{\rho_0} \sqrt{\rho_1})] & : \text{Quantum Bhattachayya error exponent} \\ \xi_{QC} &= -\log[\min_{0 \leq \alpha \leq 1} \text{tr}(\rho_0^\alpha \rho_1^{1-\alpha})] & : \text{Quantum Chernoff error exponent} \end{aligned}$$

Practical receiver

- OPA(optical parametric amplifier)



$$\hat{c}_{(k)} = \sqrt{G}\hat{a}_I^{(k)} + \sqrt{G-1}\hat{a}_R^{\dagger(k)} \quad \hat{d}_{(k)} = \sqrt{G}\hat{a}_R^{(k)} + \sqrt{G-1}\hat{a}_I^{\dagger(k)}$$

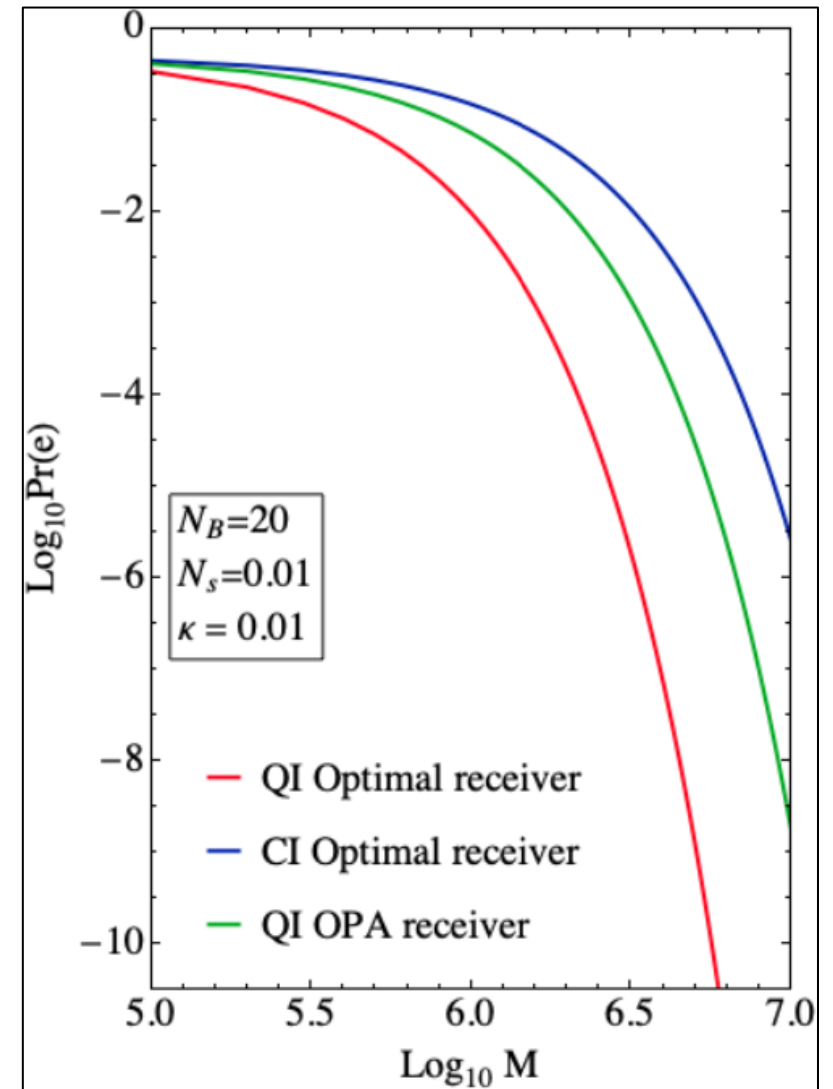
$$N_k = \langle \hat{c}_{(k)}^\dagger \hat{c}_{(k)} \rangle$$

$$N_0 = GN_S + (G-1)(1+N_B)$$

$$N_1 = GN_S + (G-1)(1+N_B + \kappa N_S) + 2\sqrt{G(G-1)}\sqrt{\kappa N_S(N_S+1)}$$

$$P_{e,OPA} \leq \frac{1}{2} Q_B^M$$

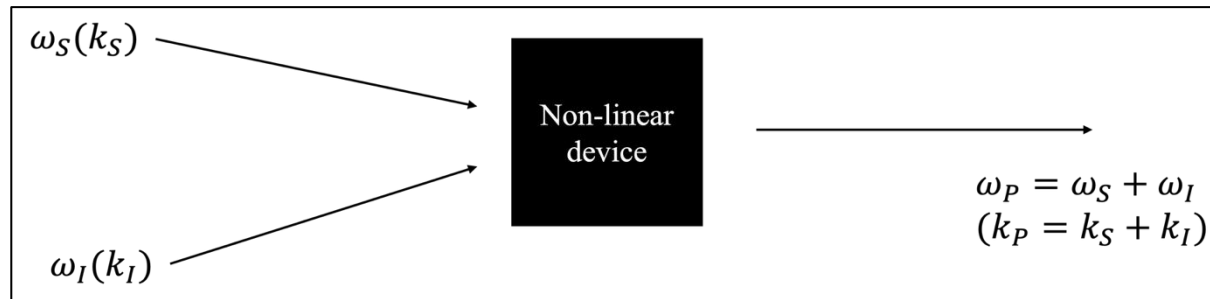
$$Q_B = \frac{1}{\sqrt{(1+N_1)(1+N_0)} - \sqrt{N_0 N_1}}$$



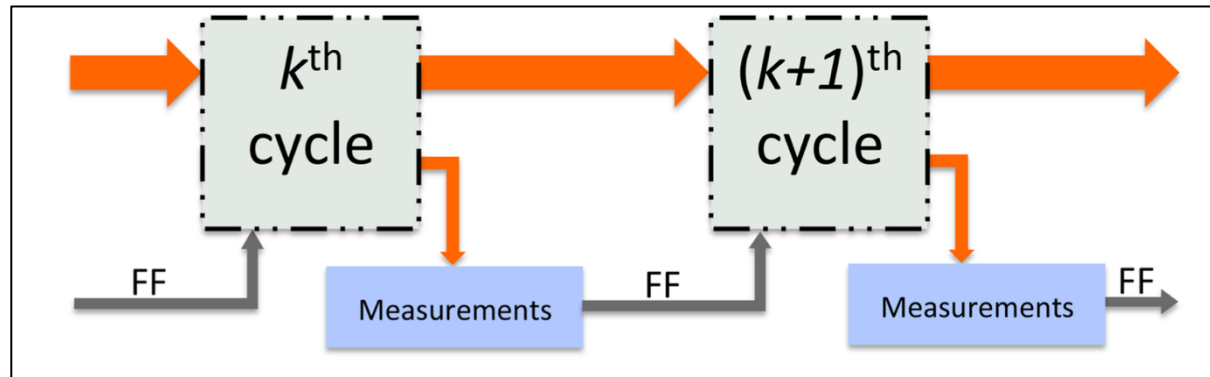
Practical receiver

- FF-SFG receiver

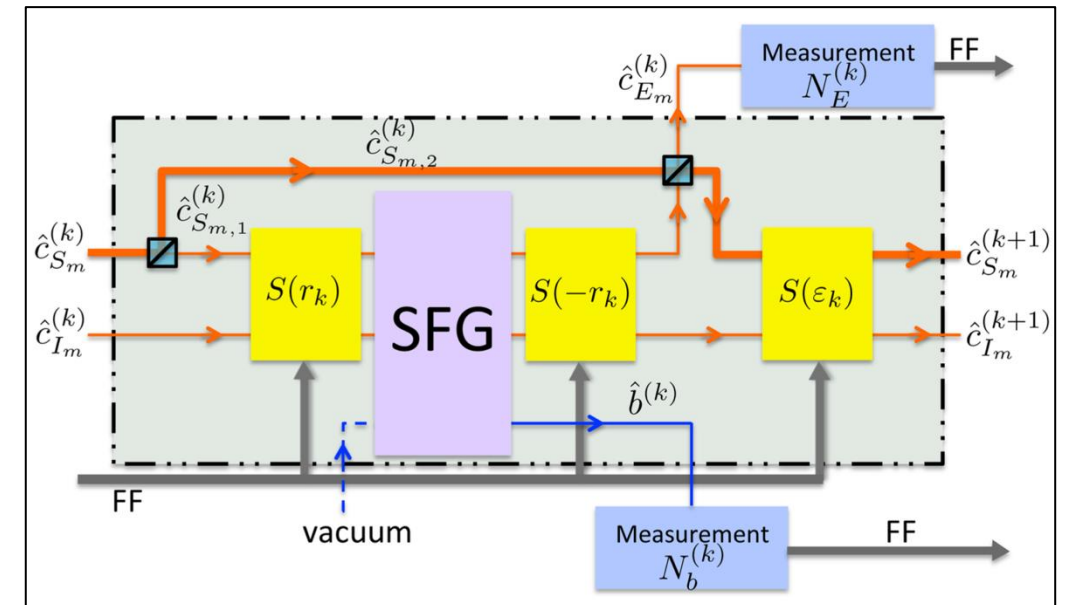
SFG(sum frequency generation)



FF(feed-forward) circuit



FF-SFG(k -th cycle)



- Saturate the Chernoff bound
- Unit efficiency at single pair level
- $|0\rangle$ (H_0 is true), $|N_S \kappa M / N_B\rangle$ (H_1 is true)

Performance : ROC

- ROC for quantum illumination

Neyman-Pearson hypothesis testing

$$P_F = \text{tr}(E_1 \rho_0) = \alpha \text{ (fixed)} \quad P_M = \text{tr}(E_0 \rho_1)$$

$$\Rightarrow P_D = 1 - P_M = \text{tr}((\mathbb{I} - E_0) \rho_1) = \text{tr}(E_1 \rho_1)$$

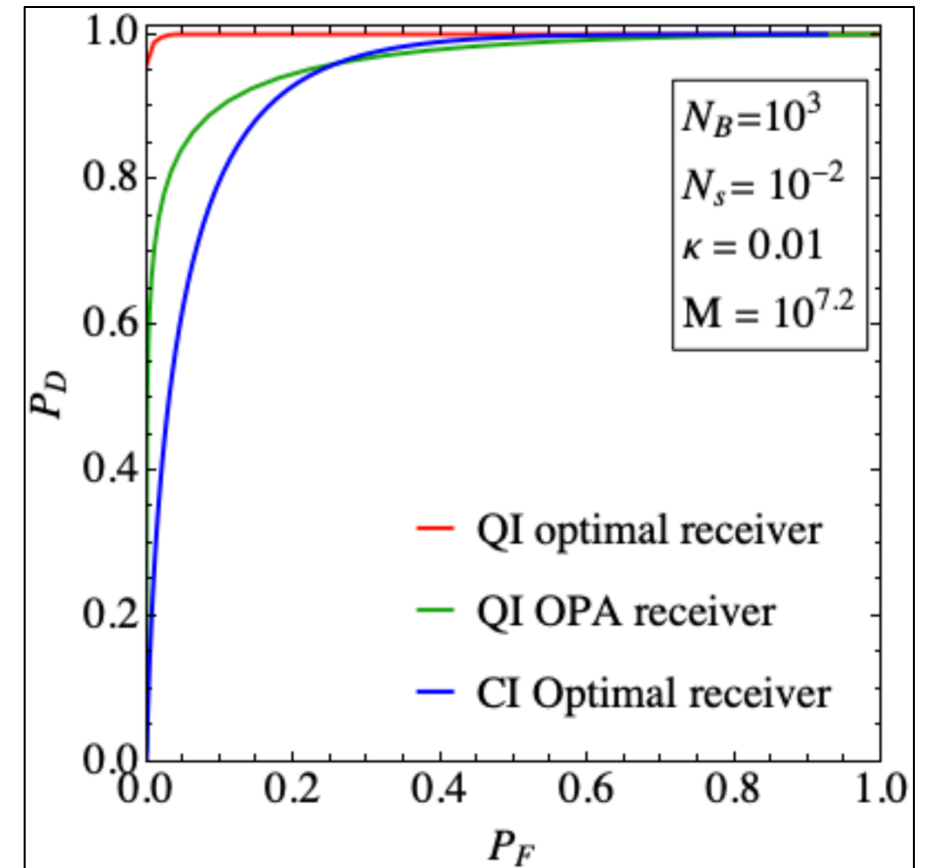
$$\rho_0 = |\psi_0\rangle\langle\psi_0| \quad \rho_1 = |\psi_1\rangle\langle\psi_1| \text{ (pure states)}$$

$$P_D = \begin{cases} (\sqrt{P_F(1-h)} + \sqrt{(1-P_F)h})^2 & 0 \leq P_F \leq 1-h \\ 1 & 1-h \leq P_F \leq 1 \end{cases}$$

- QI with the FF-SFG receiver

$$P_D = \begin{cases} (\sqrt{P_F(1-h)} + \sqrt{(1-P_F)h})^2 & 0 \leq P_F \leq 1-h \\ 1 & 1-h \leq P_F \leq 1 \end{cases}$$

$$h = 1 - \exp(-N_s \kappa M / N_B)$$



Normal mode decomposition of Gaussian states

- Normal mode decomposition of Gaussian states $(\bar{\mathbf{x}}, \mathbf{S}, \{\nu_k\})$

Every Covariance matrix $\mathbf{V}$ exists a symplectic matrix $\mathbf{S}$

$$\mathbf{V} = \mathbf{S} \left[\bigoplus_{k=1}^n \nu_k \mathbf{I}_k \right] \mathbf{S}^T \quad \leftrightarrow \quad \rho = \hat{U}_{\bar{\mathbf{x}}, \mathbf{S}} \left[\bigotimes_{k=1}^n \sigma(\nu_k) \right] \hat{U}_{\bar{\mathbf{x}}, \mathbf{S}}^\dagger$$

: Normal mode decomposition of the Gaussian state

$$\begin{pmatrix} \nu_1 & & & \\ & \nu_1 & & \\ & & \ddots & \\ & & & \nu_n \\ & & & & \nu_n \end{pmatrix}$$

(symplectic diagonalization)

$\{\nu_1, \dots, \nu_n\}$: symplectic spectrum

$$= \frac{2}{\nu_k + 1} \sum_{j=0}^{\sigma(\nu_k)} \left(\frac{\nu_k - 1}{\nu_k + 1} \right)^j |j\rangle_k \langle j|$$

: Thermal state

Two basic functions

$$\Lambda_s(x) = \frac{(x+1)^s + (x-1)^s}{(x+1)^s - (x-1)^s}$$

$$G_s(x) = \frac{2^s}{(x+1)^s - (x-1)^s}$$

Normal mode decomposition of Gaussian states

- Consider two arbitrary n -mode Gaussian state ρ_A and ρ_B with a normal mode decompositions $(\bar{\mathbf{x}}_A, \mathbf{S}_A, \{\alpha_k\})$ and $(\bar{\mathbf{x}}_B, \mathbf{S}_B, \{\beta_k\})$

$$\mathbf{d} = \bar{\mathbf{x}}_A - \bar{\mathbf{x}}_B$$

$$Q_s = \text{Tr}(\rho_A^s \rho_B^{1-s}) \quad \leftrightarrow \quad Q_s = \bar{Q}_s \exp\left\{-\frac{1}{2} \mathbf{d}^T [\mathbf{V}_A(s) + \mathbf{V}_B(1-s)]^{-1} \mathbf{d}\right\}$$

$$\bar{Q}_s = 2^n \prod_{k=1}^n G_s(\alpha_k) G_{1-s}(\beta_k) / \sqrt{\det[\mathbf{V}_A(s) + \mathbf{V}_B(1-s)]}$$

$$\mathbf{V}_A(s) = \mathbf{S}_A [\oplus_{k=1}^n \Lambda_s(\alpha_k) \mathbf{I}_k] (\mathbf{S}_A)^T$$

$$\mathbf{V}_B(1-s) = \mathbf{S}_B [\oplus_{k=1}^n \Lambda_{1-s}(\beta_k) \mathbf{I}_k] (\mathbf{S}_B)^T$$

$$P^{(N)}_{QC} = \frac{1}{2} [\inf_{0 \leq s \leq 1} Q_s]^{(N)} \quad : \text{Chernoff bound}$$

$$P^{(N)}_{QB} = \frac{1}{2} [Q_{s=1/2}]^{(N)} \quad : \text{Bhattacharyya bound}$$

Reconstruction of Classical illumination

- Calculate Bhattacharyya bound (for the Coherent-state transmitter)

Mean vector

$$\bar{\mathbf{x}}_0 = (0, 0) \quad \bar{\mathbf{x}}_1 = (\sqrt{\kappa N_S}, 0)$$

Symplectic matrix

$$\mathbf{S}_0 = \mathbf{S}_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Symplectic spectra (eigenvalue)

$$\nu_1 = B = 2N_B + 1$$

Covariance matrix

$$V_0 = V_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \Lambda(\nu_1) & 0 \\ 0 & \Lambda(\nu_1) \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

```

In[1]:= v = (2 * Nb + 1); S = IdentityMatrix[2]; s = 1 / 2;
G[x_] := 2 ^ s / ((x + 1) ^ s - (x - 1) ^ s);  $\Lambda[x_] := ((x + 1) ^ s + (x - 1) ^ s) / ((x + 1) ^ s - (x - 1) ^ s)$ ;
X0 = {0, 0}; X1 = {(2 * (k * Ns) ^ (1 / 2)), 0}; d = X0 - X1;
V0 = S.DiagonalMatrix[{ $\Lambda[v]$ ,  $\Lambda[v]$ }] . S; V1 = S.DiagonalMatrix[{ $\Lambda[v]$ ,  $\Lambda[v]$ }] . S;
Qsbar = (2 * G[v] ^ 2) / (Det[V0 + V1] ^ (1 / 2));
Qs = Qsbar * Exp[(-1 / 2) * d.Inverse[V0 + V1].d];
Qs // FullSimplify
P[M_] := (1 / 2) * (Qs) ^ M
P[M] // FullSimplify

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$$\text{Out[6]} = \frac{e^{(-1-2Nb+2\sqrt{Nb}\sqrt{1+Nb})Ns\kappa}}{(\sqrt{Nb}-\sqrt{1+Nb})^2\sqrt{1+8Nb+8Nb^2+4\sqrt{Nb}\sqrt{1+Nb}+8Nb^{3/2}\sqrt{1+Nb}}}$$

$$\text{Out[8]} = \frac{1}{2} \left(\frac{e^{(-1-2Nb+2\sqrt{Nb}\sqrt{1+Nb})Ns\kappa}}{(\sqrt{Nb}-\sqrt{1+Nb})^2\sqrt{1+8Nb+8Nb^2+4\sqrt{Nb}\sqrt{1+Nb}+8Nb^{3/2}\sqrt{1+Nb}}} \right)^M$$

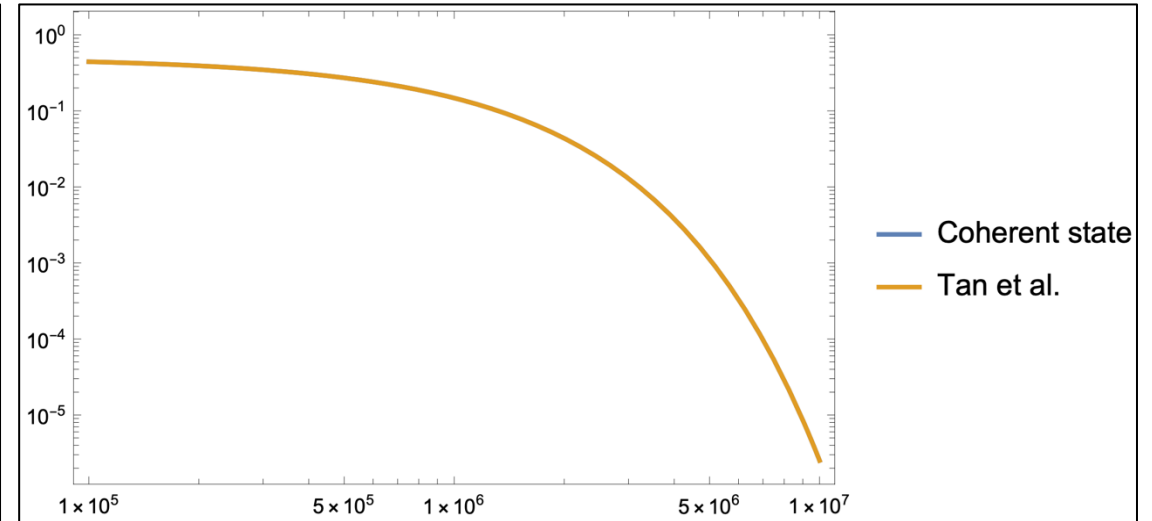
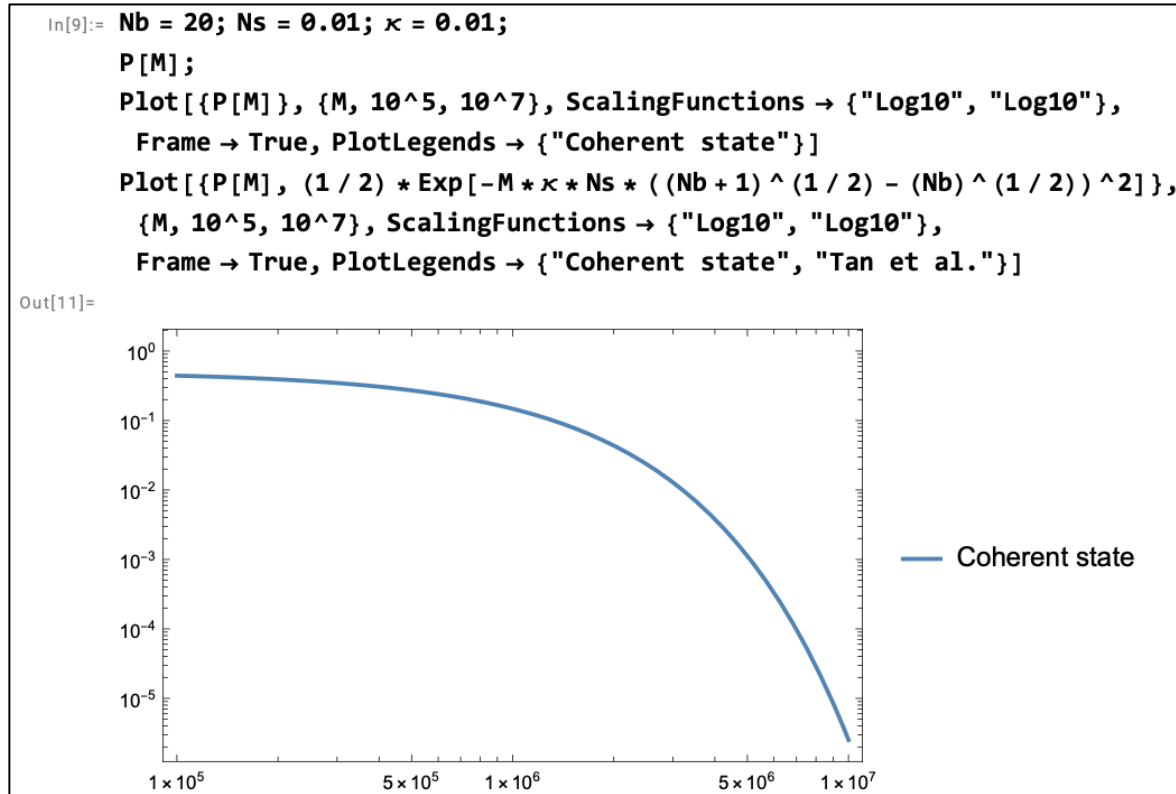
Analytic expression for Bhattacharyya bound

$$\Pr(e)_{\text{CS}} \leq e^{-M\kappa N_S(\sqrt{N_B+1}-\sqrt{N_B})^2/2}$$

Reconstruction of Classical illumination

- Calculate Bhattacharyya bound (for the Coherent-state transmitter)

Upper bounds on the target-detection error probabilities for coherent state



- $N_B = 20, \kappa = 0.01, N_s = 0.01$
- Coincide with the Chernoff bounds

Reconstruction of Quantum illumination

- Calculate Bhattacharyya bound (for the TMSV transmitter)

Mean vector $\bar{\mathbf{x}}_0 = \bar{\mathbf{x}}_1 = (0, 0, 0, 0)$

Symplectic matrix

$$\mathbf{S}_0 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \mathbf{S}_1 = \begin{pmatrix} \mathbf{X}_+ & \mathbf{X}_- \\ \mathbf{X}_- & \mathbf{X}_+ \end{pmatrix}$$

$$\mathbf{X}_{\pm} = \begin{pmatrix} x_{\pm} & 0 \\ 0 & x_{\pm} \end{pmatrix} \quad x_{\pm} = \sqrt{\frac{A + S \pm \sqrt{(A + S)^2 - 4\kappa C_q^2}}{2\sqrt{(A + S)^2 - 4\kappa C_q^2}}}$$

Symplectic spectra (eigenvalue)

$$\nu_{0_1} = B = 2N_B + 1, \nu_{0_2} = S = 2N_S + 1$$

$$\nu_{1_k} = \frac{1}{2} \left[(-1)^k (S - A) + \sqrt{(A + S)^2 - 4\kappa C_q^2} \right] \quad (k = 1, 2)$$

Covariance matrix

$$V_0 = \mathbf{S}_0 \begin{pmatrix} \Lambda(\nu_{0_1}) & 0 & 0 & 0 \\ 0 & \Lambda(\nu_{0_1}) & 0 & 0 \\ 0 & 0 & \Lambda(\nu_{0_2}) & 0 \\ 0 & 0 & 0 & \Lambda(\nu_{0_2}) \end{pmatrix} \mathbf{S}_0^T$$

$$V_1 = \mathbf{S}_1 \begin{pmatrix} \Lambda(\nu_{1_1}) & 0 & 0 & 0 \\ 0 & \Lambda(\nu_{1_1}) & 0 & 0 \\ 0 & 0 & \Lambda(\nu_{1_2}) & 0 \\ 0 & 0 & 0 & \Lambda(\nu_{1_2}) \end{pmatrix} \mathbf{S}_1^T$$

Reconstruction of Quantum illumination

- Calculate Bhattacharyya bound (for the TMSV transmitter)

```

In[1]:= S = 1 / 2;
        S = 2 * Ns + 1; B = 2 * Nb + 1; A = 2 * κ * Ns + B; Cq = 2 * Sqrt[Ns * (Ns + 1)];
        G[x_] := 2^S / ((x + 1)^S - (x - 1)^S);
        Δ[x_] := ((x + 1)^S + (x - 1)^S) / ((x + 1)^S - (x - 1)^S);
        Xplus = Sqrt[(A + S + Sqrt[(A + S)^2 - 4 * κ * Cq^2]) / (2 * Sqrt[(A + S)^2 - 4 * κ * Cq^2])];
        Xminus = Sqrt[(A + S - Sqrt[(A + S)^2 - 4 * κ * Cq^2]) / (2 * Sqrt[(A + S)^2 - 4 * κ * Cq^2])];
        V = IdentityMatrix[4];
        U = {{Xplus, 0, Xminus, 0}, {0, Xplus, 0, -Xminus},
             {Xminus, 0, Xplus, 0}, {0, -Xminus, 0, Xplus}};
        v1 = B; v2 = S; (*under H0*)
        v3 = (1 / 2) * ((A - S) + Sqrt[(A + S)^2 - 4 * κ * (Cq)^2]);
        v4 = (1 / 2) * ((S - A) + Sqrt[(A + S)^2 - 4 * κ * (Cq)^2]); (*under H1*)
        V0 = V.DiagonalMatrix[{Δ[v1], Δ[v1], Δ[v2], Δ[v2]}].V;
        V1 = U.DiagonalMatrix[{Δ[v3], Δ[v3], Δ[v4], Δ[v4]}].Transpose[U];
        Qs = 4 * (G[v1] * G[v3] * G[v2] * G[v4]) / Sqrt[Det[V0 + V1]];
        P[M_] := (1 / 2) * (Qs)^M
        P[M];
    
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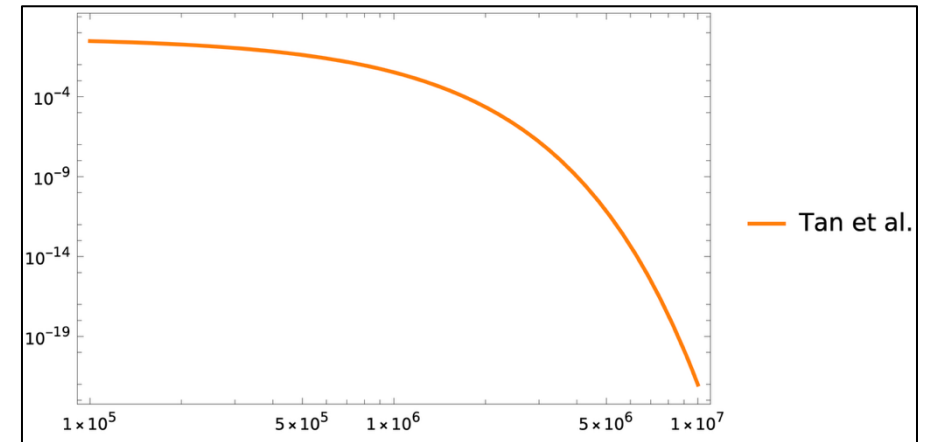
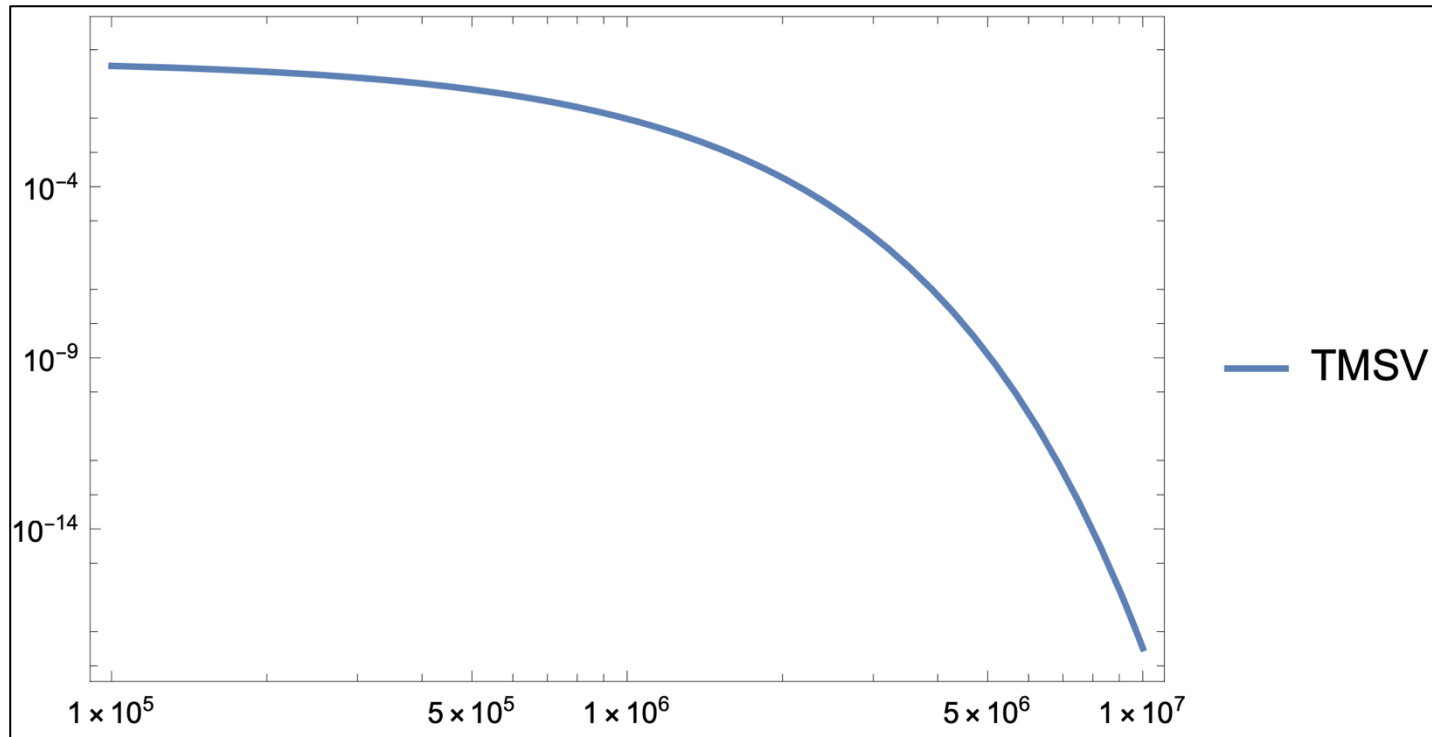
$$\begin{aligned}
 \text{Out[11]} = & 16 / \left(\left(-\sqrt{2} \sqrt{Nb} + \sqrt{2+2Nb} \right) \left(-\sqrt{2} \sqrt{Ns} + \sqrt{2+2Ns} \right) \right. \\
 & \left(-\sqrt{-1 + \frac{1}{2} \left(-2Nb + 2Ns - 2Ns\kappa + \sqrt{-16Ns(1+Ns)\kappa + (2+2Nb+2Ns+2Ns\kappa)^2} \right)} + \right. \\
 & \left. \sqrt{1 + \frac{1}{2} \left(-2Nb + 2Ns - 2Ns\kappa + \sqrt{-16Ns(1+Ns)\kappa + (2+2Nb+2Ns+2Ns\kappa)^2} \right)} \right) \\
 & \left(-\sqrt{-1 + \frac{1}{2} \left(2Nb - 2Ns + 2Ns\kappa + \sqrt{-16Ns(1+Ns)\kappa + (2+2Nb+2Ns+2Ns\kappa)^2} \right)} + \right. \\
 & \left. \sqrt{1 + \frac{1}{2} \left(2Nb - 2Ns + 2Ns\kappa + \sqrt{-16Ns(1+Ns)\kappa + (2+2Nb+2Ns+2Ns\kappa)^2} \right)} \right) \\
 & \left. \right) \\
 & \vdots \\
 \text{Out[13]} = & 2^{-1+4M} \left(1 / \left(\left(-\sqrt{2} \sqrt{Nb} + \sqrt{2+2Nb} \right) \left(-\sqrt{2} \sqrt{Ns} + \sqrt{2+2Ns} \right) \right. \right. \\
 & \left(-\sqrt{-1 + \frac{1}{2} \left(-2Nb + 2Ns - 2Ns\kappa + \sqrt{-16Ns(1+Ns)\kappa + (2+2Nb+2Ns+2Ns\kappa)^2} \right)} + \right. \\
 & \left. \sqrt{1 + \frac{1}{2} \left(-2Nb + 2Ns - 2Ns\kappa + \sqrt{-16Ns(1+Ns)\kappa + (2+2Nb+2Ns+2Ns\kappa)^2} \right)} \right) \\
 & \left. \left. \right) \right) \\
 & \vdots
 \end{aligned}$$

- Unable to obtain an analytic expression of compact form (too long)
- Fail to simplify (exceed the time limit 300s)

Reconstruction of Quantum illumination

- Calculate Bhattacharyya bound (for the TMSV transmitter)

Upper bounds on the target-detection error probabilities for TMSV



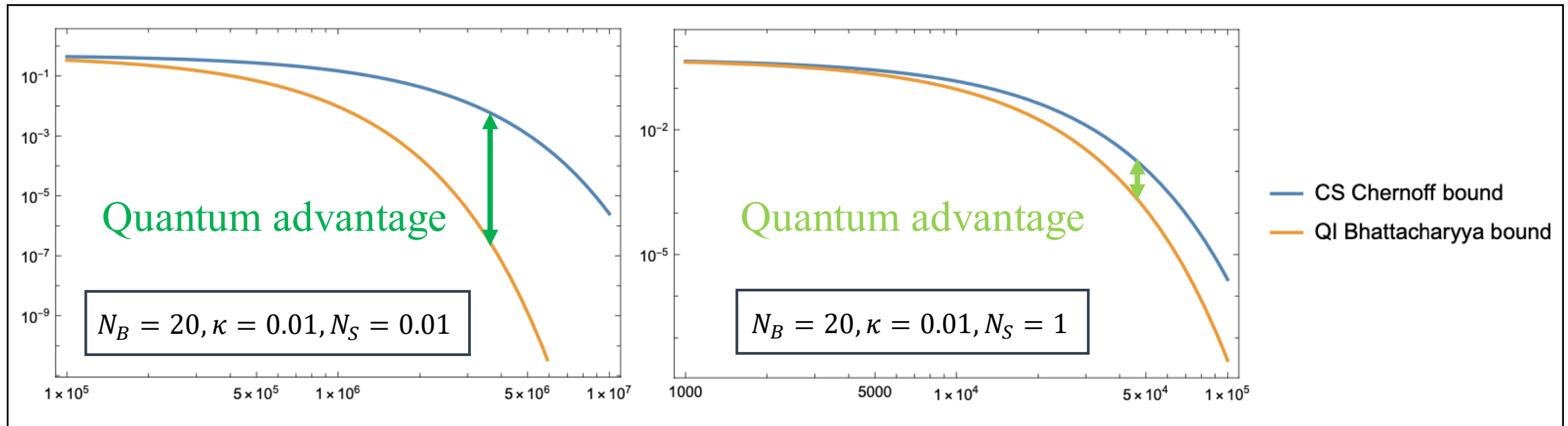
$$\Pr(e)_{\text{QI}} \leq e^{-M\kappa N_S/N_B}/2$$

$$\kappa \ll 1, N_S \ll 1, N_B \gg 1$$

Results

- Quantum advantage becomes less and less important when increasing N_S

Upper bounds on the target-detection error probabilities for TMSV

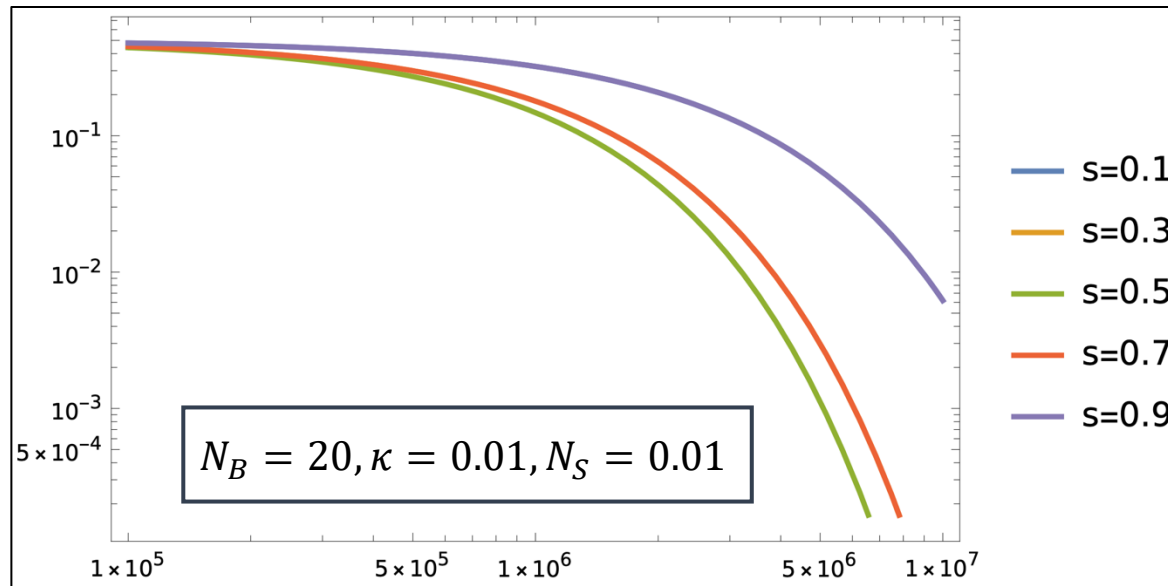


- Show the clear quantum advantage in the $N_S \ll 1$
- Quantum advantage is significantly depleted in the $N_S = 1$

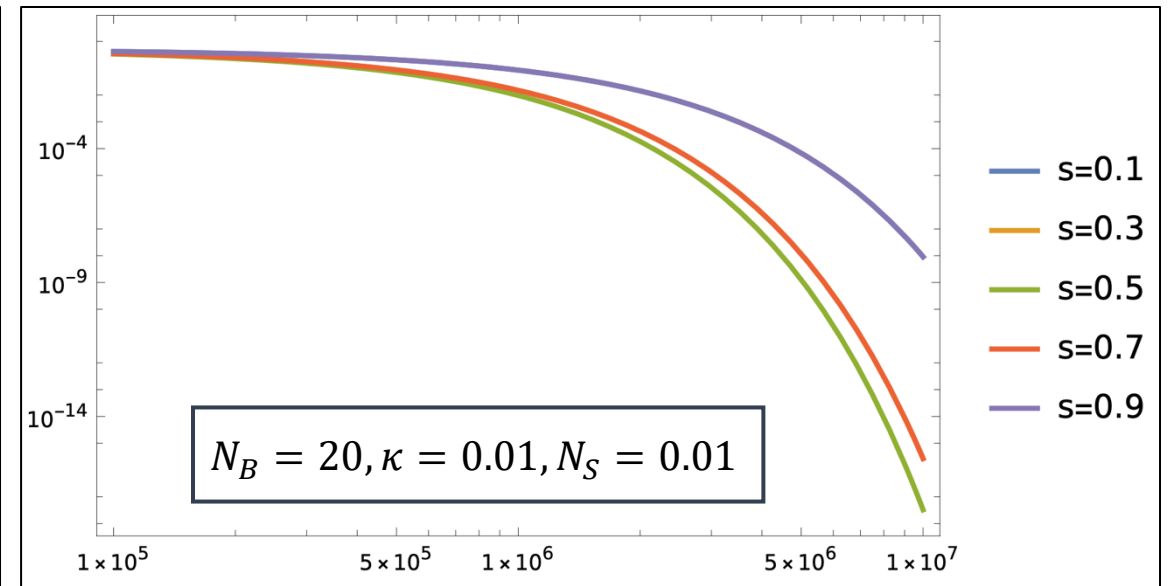
Results

- Target-detection error probabilities according to s (optimization over s)

Upper bounds on the P_e for the CI



Upper bounds on the P_e for the QI



- Symmetrical with respect to $s = 0.5$
- Lowest target-detection error probability at $s = 0.5$
- However, Chernoff bound $\neq$ Bhattacharyya bound in the QI (for the TMSV transmitter)

Discussion & Further research

- Reconstruction of Classical illumination
 - Find more simplified analytic expression by using approximation ($\kappa \ll 1, N_S \ll 1, N_B \gg 1$)
 - Check to have Chernoff bound at $s = 0.5$ by numerical method (implement Mathematica code)
 - Reconstruction of Quantum illumination
 - Find analytic solution of compact form by using approximation ($\kappa \ll 1, N_S \ll 1, N_B \gg 1$)
 - Calculate quantum Chernoff bound by optimization over s
-
- Further research
 - Calculate target detection error probability for other Gaussian states
 - Calculate ROC for the coherent-state transmitter, TMSV transmitter
 - Implement illumination protocol using practical receiver (OPA, SS-SFG)

Reference & Acknowledgment

- Reference

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Thank you for your attention!